MLESA-Net: Attention-Enhanced Multi-Label Emotion and Sentiment Analysis on Nepali Reddit Corpus

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Abstract. Emotion and sentiment analysis of low-resource, code-mixed languages such as Nepali presents significant challenges owing to limited annotated corpora, morphological complexity, and mixed-script social media content. We proposed a Multi-Head Attention framework named **MLESA-Net** based on deep learning models such as BiLSTM, CNN-LSTM and multilingual transformers such as XLM-RoBERTa is evaluated on two Nepali Reddit corpora. We have evaluated the proposed architecture on the benchmark datasets such as NepEMO, MELD, GoEmotions and our newly introduced dataset named **NEmoSen** corpora. On NepEMO, our proposed model **MLESA-Net** achieves the best emotion scores such as Mic-F1 is **0.6991** and Mic-F1 is **0.6851**, while XLM-RoBERTa performs better on the GoEmotions dataset having 0.462 and Mic-F1 as 0.554. The performance of mBERT is comparatively better on the MELD data, with Mic-F1 and Mac-F1 scores as 0.347 and 0.503 respectively. On our smaller **NEmoSen** dataset, Linear SVM outperforms all deep learning and transformer models (Micro-F1 = 0.6309), confirming that large pretrained models require proportionally larger training corpora to generalise effectively on code mixed low-resource text.

Keywords: Emotion Analysis · Sentiment Analysis · Nepali NLP · Attention Mechanism · Code-Mixed Language · XLM-RoBERTa · mBERT

1 Introduction

The growth of social-media platforms has created an unprecedented opportunity to study human emotion at scale. Posts on Nepali Reddit communities openly express complex emotional states in a highly code-mixed register—combining Romanised Nepali, Devanagari script, and English in the same sentence—making standard NLP pipelines inadequate without specialised preprocessing and multilingual modelling.

Emotion analysis is inherently a *multi-label* problem: a single post may simultaneously express fear, sadness, and depression. This paper investigates

three central research questions: **RQ1**: Can attention mechanisms bridge the performance gap between base deep learning models and transformers on code-mixed Nepali text? **RQ2**: How does training-corpus scale determine optimal model selection in low-resource settings? **RQ3**: Which emotion and sentiment classes remain persistently difficult across all architectures, and why?

Contributions.

1. Systematic benchmark of ten diverse models on Nepali multi-label emotion and multi-class sentiment classification across four datasets (NepEMO, NEmoSen, MELD, GoEmotions).
2. Three attention-enhanced architectures (BiLSTM + Simple Attention, BiLSTM + Multi-Head Attention, CNN-LSTM + Attention) evaluated on code-mixed Nepali text, with the MHA variant yielding a +21.0 % gain on NepEMO.
3. A new NEmoSen corpus (1,059 samples, 2025–2026) from `r/Nepal` with manual annotation following the NepEMO schema, publicly released at https://github.com/Nabin68/NEmoSen-Nepali-Emotion-Sentiment-Corpus-.
4. Empirical evidence that for corpora below ≈750 training samples, TF-IDF + SVM consistently dominates transformers, providing a practical data-size heuristic for low-resource NLP.

2 Related Work

2.1 Emotion and Sentiment Analysis

Sentiment analysis has been extensively studied for English, with CNN-based models [7] and BERT [3] achieving strong results. Multi-label emotion classification requires models that handle label co-occurrence and class imbalance simultaneously. Attention mechanisms [6,5] have consistently improved sequence-model performance for affective computing tasks. The MELD dataset [12] and GoEmotions [13] provide established multi-class emotion benchmarks that we use for cross-lingual generalisation evaluation in this work.

2.2 Low-Resource and Code-Mixed NLP

XLM-RoBERTa [4] (pretrained on 100 languages) provides strong cross-lingual transfer for low-resource scenarios. Multilingual BERT [3] has demonstrated utility for code-mixed settings. Code-mixing in South Asian languages, including Nepali, poses challenges that monolingual models cannot adequately address due to the heterogeneous script and vocabulary distributions.

2.3 Nepali NLP

Nepali NLP has seen growing interest, yet remains severely under-resourced. Shahi and Sitaula [15] provide a comprehensive survey of Nepali NLP resources and tasks, documenting progress across classification, sentiment, and morphological

Table 1. Emotion-Aligned Keyword Groups for Data Collection

Emotion	English	Romanised Nepali
Sadness	sad, depressed, heartbroken, alone, breakup	dukha, dukhi, roye, eklo
Anger	angry, frustrated, hate	ris, risa, jhyau, hairan, wakka
Fear	scared, anxiety	dar, tension, chinta, darr lagyo
Joy	happy, excited, proud, amazing	khusi, ramailo, maja, jeetyo
Surprise	shocked, unexpected, unbelievable	achamma, wow

analysis. Sitaula et al. [16] applied deep learning to sentiment analysis of Nepali COVID-19-related social-media posts, demonstrating the feasibility of neural approaches on short, noisy text. The NepEMO dataset [1] is the first large-scale multi-label emotion corpus for Nepali Reddit, annotated with five emotion categories (Fear, Anger, Sadness, Joy, Depression) and three sentiment classes, and serves as the primary benchmark for this study.

3 Datasets

3.1 NepEMO Dataset

The NepEMO benchmark [1] contains 4,462 Nepali Reddit posts with five binary emotion labels and a three-class sentiment label (Positive, Negative, Neutral). Text is code-mixed (English, Romanised Nepali, Devanagari), with mean word count 118.1, median 55 words, and an average of 1.62 simultaneous emotion labels per post.

3.2 NEmoSen: Nepali Emotion & Sentiment Corpus

To provide a temporally recent complement, we constructed an original dataset from `r/Nepal` via the public Reddit JSON API (January 2025–February 2026), which we name **NEmoSen** (Nepali Emotion & Sentiment Corpus) [2].

Collection. Queries were organised into five emotion-aligned keyword groups combining English and Romanised Nepali terms (Table 1). For each group, up to 400 posts were retrieved (`restrict_sr=1`, `sort=new`). Both post bodies (*title*+*selftext*) and top-level comments were collected; each comment is treated as an independent sample. A 1 s delay between comment requests and 2 s between pagination calls was applied per Reddit API rate-limit guidelines. Duplicates were removed via exact-string matching; samples with fewer than 8 words post-cleaning were discarded.

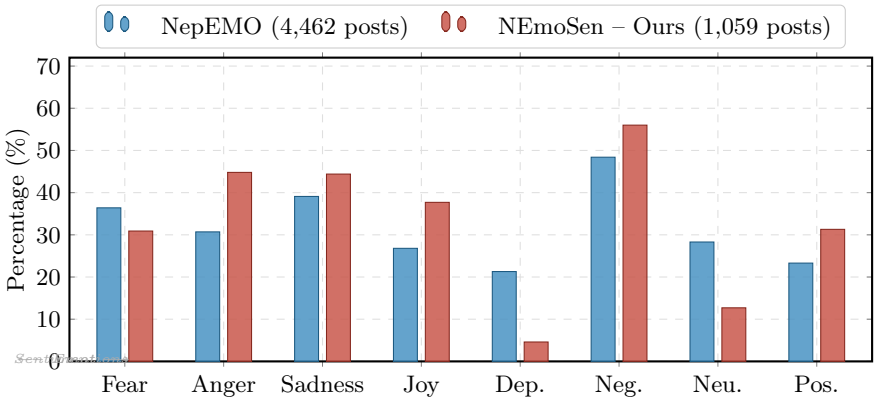


Fig. 1. Emotion and sentiment label percentages for NepEMO and the NEmoSen corpus. NEmoSen shows notably lower Depression (4.6 % vs. 21.3 %) and higher Anger/Sadness proportions, reflecting recent discourse shifts in `r/Nepal`.

Annotation. The 1,059-sample corpus was divided into two non-overlapping partitions; each author independently annotated their respective partition following the NepEMO schema: five binary emotion labels (*Joy, Anger, Fear, Sadness, Depression*) and a three-class sentiment label (0 = Neutral, 1 = Positive, 2 = Negative), permitting multiple emotion labels per sample. Both authors jointly reviewed samples where confidence was low, resolving disagreements through discussion. The final dataset comprises **1,059 unique samples** and is publicly released to facilitate reproducibility and future benchmarking at https://github.com/Nabin68/NEmoSen-Nepali-Emotion-Sentiment-Corpus-.

3.3 Cross-Dataset Generalisation: MELD and GoEmotions

To assess how models trained on Nepali data generalise beyond their training domain, we additionally evaluate all ten models on two widely-used English emotion benchmarks. MELD [12] is a multimodal, multi-party conversational emotion dataset with seven emotion classes (1,039 training dialogues). GoEmotions [13] is a fine-grained emotion dataset of 58,000 English Reddit comments spanning 28 emotion categories. For both benchmarks, each model is trained from scratch using the respective official train/validation/test splits with the same preprocessing and training pipeline described in Section 4.

3.4 Dataset Comparison

Table 2 presents a side-by-side statistical comparison of the two Nepali corpora; Fig. 1 visualises the label distributions.

Table 2. NepEMO vs. NEmoSen: Side-by-Side Comparison

Attribute	**NepEMO [1]**		**NEmoSen [2] (Ours)**	
	Count	%	Count	%
Total samples	4,462	–	1,059	–
Period	pre-2024		Jan 2025–Feb 2026	
Avg. words/post	118.1		80.8	
Median words	55		47	
Avg. labels/post	1.62		1.62	
Emotion labels				
Fear	1,625	36.4	327	30.9
Anger	1,369	30.7	474	44.8
Sadness	1,744	39.1	470	44.4
Joy	1,195	26.8	399	37.7
Depression	949	21.3	49	4.6
Sentiment labels				
Negative	2,161	48.4	593	56.0
Neutral	1,261	28.3	135	12.7
Positive	1,040	23.3	331	31.3
Train / Val / Test split (70 / 15 / 15)				
Train	3,123	70	741	70
Validation	670	15	159	15
Test	669	15	159	15

4 Methodology

4.1 System Architecture

Figure 2 details the complete end-to-end pipeline, mapping the flow from raw mixed-script inputs through preprocessing, feature extraction, and four parallel model tracks to dual output heads.

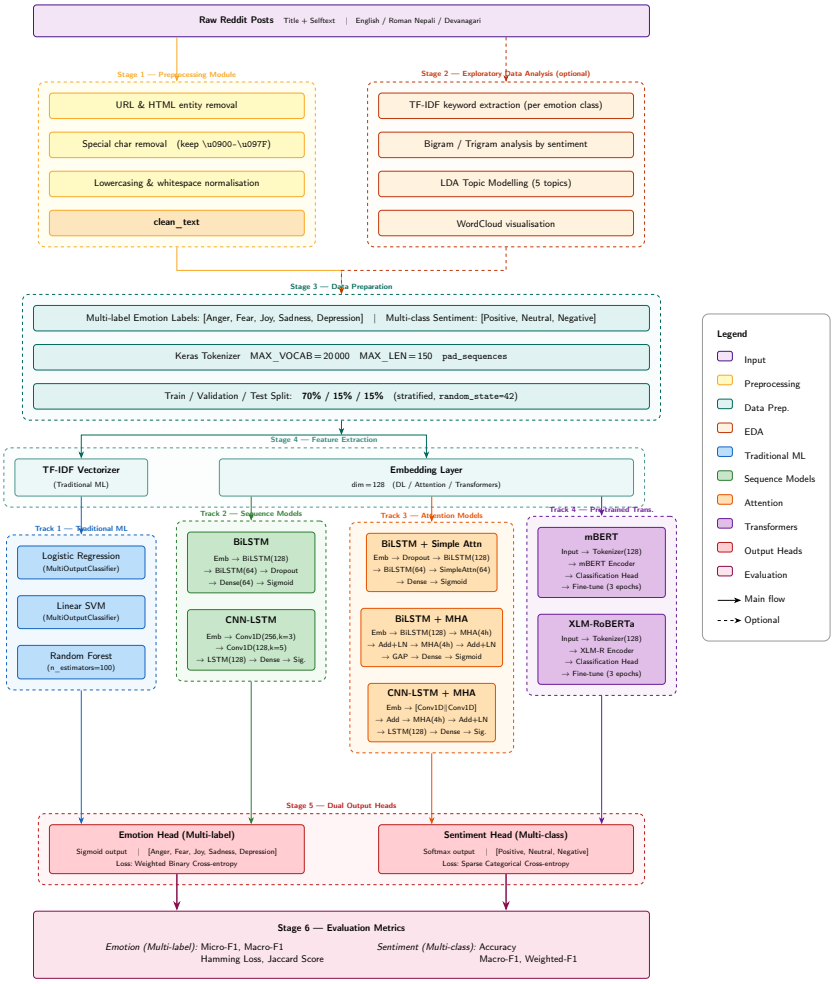


Fig. 2. End-to-end pipeline for multi-label emotion and sentiment analysis of code-mixed Nepali Reddit posts. Raw posts flow through preprocessing and data preparation before branching into four parallel model tracks, each feeding dual output heads evaluated by a common set of metrics.

4.2 Text Preprocessing

All text was preprocessed through a multilingual pipeline: (1) URL and Reddit-mention removal (`u/`, `r/`); (2) HTML entity stripping; (3) script-aware character filtering preserving Devanagari Unicode (U+0900–U+097F); (4) lowercasing of the Latin-script portion; and (5) for topic modelling, NLTK English stopwords combined with curated Romanised Nepali filler words (*ani, lai, pani, tara, chai, haru, ma, ko, ra, ni*). For the NepEMO dataset, `title` and `selftext` fields were concatenated before preprocessing.

4.3 Feature Engineering

TF-IDF vectorisation used 10,000 features, (1,2)-gram range, sublinear TF scaling, and minimum document frequency of 2, fitted exclusively on the training split to prevent data leakage. Keras deep learning models used a 20,000-word vocabulary tokeniser with an `<OOV>` token, post-padded to 150 tokens. Transformer models used their respective `AutoTokenizer` with a maximum sequence length of 128 tokens.

4.4 Attention Mechanisms

Figure 3 illustrates the two attention architectures proposed in this work, alongside Figure 4 evaluating their performance relative to baseline models.

BiLSTM + Simple Attention [6]. After two stacked BiLSTM layers [9,8] (128 and 64 units), a feed-forward scoring network computes a scalar importance weight α_t for each hidden state h_t: $e_t = \mathbf{v}^\top \tanh(\mathbf{W}h_t)$, $\alpha_t = \mathrm{softmax}(e_t)$. The context vector $\mathbf{c} = \sum_t \alpha_t h_t$ feeds the classification head.

BiLSTM + Multi-Head Attention [5]. Two stacked Multi-Head Attention blocks (4 heads, $d_k = 32$) follow the BiLSTM output. Each block applies a residual connection and Layer Normalisation. Global Average Pooling then collapses the attended sequence before the dense classification layers.

4.5 Model Training Configuration

All models use the same 70 / 15 / 15 stratified split (`random_state=42`) applied consistently across all four datasets. Traditional ML models—Logistic Regression, Linear SVM [10], and Random Forest [11]—use `MultiOutputClassifier` wrappers for the multi-label emotion task and direct classifiers for sentiment. Deep learning models apply EarlyStopping (patience = 4) and ReduceLROnPlateau (factor = 0.5, patience = 2, min LR = 10^{-6}). CNN-LSTM variants use a custom weighted binary cross-entropy loss in which rare emotion classes receive higher penalties proportional to their inverse frequency. Transformers are fine-tuned for 3 epochs with AdamW [14] (weight decay = 0.01, 100 warm-up steps), batch size 16, and `BCEWithLogitsLoss` for multi-label emotion classification. For MELD [12] and GoEmotions [13], each model is trained from scratch using the respective official splits with the identical preprocessing and training pipeline; no knowledge is transferred from the Nepali datasets.

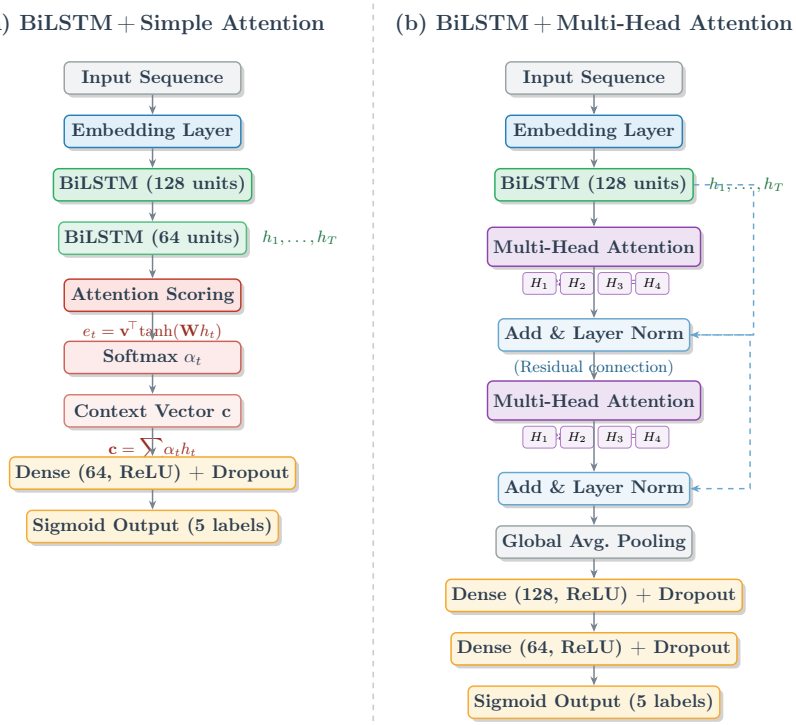


Fig. 3. Fig. 3. Proposed attention-enhanced architectures. (a) BiLSTM + Simple Attention: Bahdanau-style scorer computes per-state weights α_t; context vector $\mathbf{c} = \sum_t \alpha_t h_t$ feeds the classifier. (b) BiLSTM + Multi-Head Attention: two stacked MHA blocks (4 heads, $d_k = 32$) with residual connections (dashed) and Layer Normalisation; Global Average Pooling reduces the attended sequence.

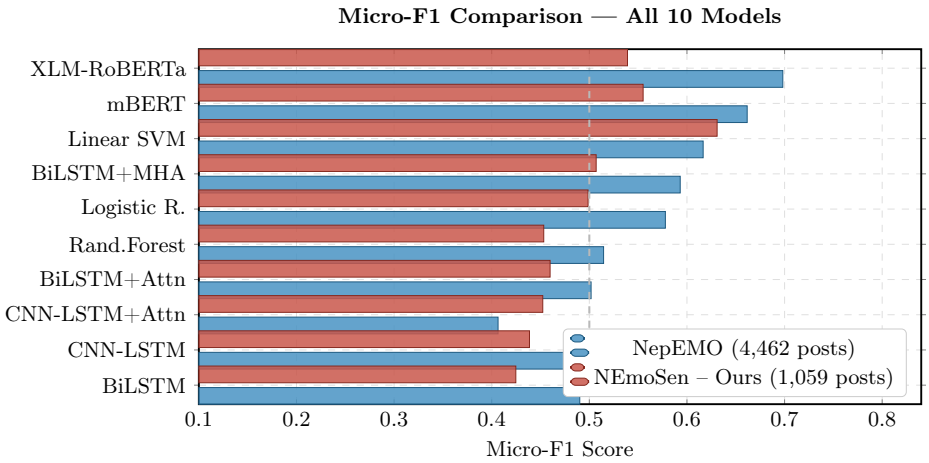


Fig. 4. Fig. 4. Micro-F1 comparison of all ten models on both Nepali datasets (ordered bottom-to-top by NepEMO score). The dashed line marks Micro-F1 = 0.50. Linear SVM leads on NEmoSen (0.6309) while XLM-RoBERTa leads on NepEMO (0.6982), driven by training-set size.

5 Experimental Results

5.1 Emotion Classification Comparison

Table 3 reports emotion classification results across all four datasets. On NepEMO, XLM-RoBERTa achieves the best Micro-F1 of 0.6982, benefiting from its 100-language pretraining corpus [4]. On the data-constrained NEmoSen dataset (741 training samples), Linear SVM leads with Micro-F1 = 0.6309 while XLM-RoBERTa declines to 0.5391—a rank reversal discussed in Section 6. Notably, BiLSTM + MHA achieves near-zero F1 on both MELD and GoEmotions (0.005 and 0.001 respectively); this collapse is explained by an architecture mismatch: its dual sigmoid heads are designed for multi-label binary prediction and are ill-suited to the mutually exclusive single-label classification required by these benchmarks.

Table 3. Emotion Classification — Micro-F1 and Macro-F1 Across All Datasets. **Bold** = best per column.

		NepEMO		NEmoSen (Ours)		MELD		GoEmotions	
Type	**Method**	**Mic-F1**	**Mac-F1**	**Mic-F1**	**Mac-F1**	**Mac-F1**	**Wtd-F1**	**Mac-F1**	**Wtd-F1**
Transformer	XLM-RoBERTa	0.6882	0.6664	0.5391	0.3641	0.093	0.313	**0.462**	**0.554**
	mBERT	0.6617	0.6533	0.5551	0.4213	**0.347**	**0.503**	0.448	0.552
ML	Linear SVM	0.6166	0.6040	**0.6309**	**0.5019**	0.292	0.464	0.348	0.474
	Logistic Regression	0.5779	0.5541	0.4987	0.3617	0.251	0.445	0.305	0.447
	Random Forest	0.5146	0.4894	0.4533	0.3362	0.251	0.442	0.297	0.431
DL+Attn	*MLESA-Net (Ours)	**0.6991**	**0.6851**	0.5070	0.3720	0.005	0.001	0.001	0.001
	BiLSTM + Simple Attn	0.5018	0.4716	0.4598	0.3200	0.248	0.289	0.332	0.323
	CNN-LSTM + Attn	0.4065	0.3152	0.4523	0.3131	0.149	0.150	0.251	0.256
DL	BiLSTM	0.4902	0.4569	0.4247	0.3336	0.260	0.458	0.306	0.317
	CNN-LSTM	0.5347	0.5306	0.4387	0.3816	0.028	0.021	0.001	0.000

NepEMO & NEmoSen: multi-label, 5 classes. MELD: single-label, 7 classes [12]. GoEmotions: single-label, 28 classes [13]. BiLSTM+MHA collapsed (near-zero F1) on MELD & GoEmotions owing to architecture mismatch: the dual sigmoid heads designed for multi-label binary prediction are ill-suited to mutually exclusive single-label classification with many classes. All models trained from scratch on each dataset's official splits. On NEmoSen, bold marks the best per column independently: Linear SVM leads emotion classification; mBERT leads among transformers.

5.2 Attention Improvement Analysis

Table 4 quantifies attention gains over base DL models. BiLSTM + MHA yields the largest absolute gain on NepEMO (+0.1029, +21.0 %) and a consistent positive gain on NEmoSen (+0.0823, +19.4 %), answering **RQ1** affirmatively: attention mechanisms do meaningfully close the gap between base DL and transformers, at substantially lower computational cost. Figure 4 provides a visual ranking of all ten models across both Nepali datasets.

Table 4. Attention Gain over Base Deep Learning Models (Micro-F1).

Base	Variant	NepEMO		NEmoSen	
		Gain	%	Gain	%
BiLSTM	+ Simple Attn	+0.0116	+2.4 %	+0.0351	+8.3 %
BiLSTM	+ MHA	+0.1029	+**21.0** %	+0.0823	+**19.4** %
CNN-LSTM	+ Attention	−0.1282	−24.0 %	+0.0136	+3.1 %

Table 5. Per-Class Emotion F1 Scores for Selected Model–Dataset Combinations. Support counts reflect the test split.

Dataset	Model	Joy	Anger	Fear	Sadness	Depression
NepEMO	XLM-RoBERTa	0.64	0.71	0.72	0.75	0.62
	BiLSTM+MHA	0.41	0.64	0.62	0.67	0.48
NEmoSen	Linear SVM	0.47	0.66	0.61	0.55	0.00
	BiLSTM	0.22	0.48	0.00	0.29	0.00

XLM-RoBERTa NepEMO per-class scores from classification_ report output (Micro-F1 = 0.6982, Macro-F1 = 0.6864, Hamming Loss = 0.1887; test support: Fear = 233, Anger = 203, Sadness = 252, Joy = 173, Depression = 146). Depression F1 = 0.00 on NEmoSen across all models due to extreme class imbalance (4.6 % of samples, ≈7 test instances).

5.3 Per-Class Emotion F1 Analysis

Table 5 presents per-emotion F1 scores for the three most informative model–dataset combinations, addressing **RQ3**.

5.4 Sentiment Classification

Table 6 shows sentiment results for traditional ML models across all datasets. Linear SVM achieves the best Macro-F1 of 0.6289 on NepEMO. On NEmoSen, all models yield low Macro-F1 (best: 0.3874) owing to near-total Neutral class failure, as discussed in Section 6.

Table 6. Sentiment Classification — Macro-F1 and Weighted-F1 Across All Datasets (Traditional ML models). **Bold** = best per column.

	NepEMO		NEmoSen (Ours)		MELD		GoEmotions	
Method	Mac-F1	Wtd-F1	Mac-F1	Wtd-F1	Mac-F1	Wtd-F1	Mac-F1	Wtd-F1
Linear SVM	**0.6289**	**0.6576**	**0.3874**	**0.5463**	0.502	0.540	0.572	0.618
Logistic Regression	0.6011	0.6349	0.3623	0.5231	**0.522**	**0.559**	**0.582**	**0.635**
Random Forest	0.5227	0.5668	0.2733	0.4315	0.477	0.525	0.540	0.602

NepEMO & NEmoSen: 3 classes (Positive/Neutral/Negative). MELD: 3 classes. GoEmotions: 4 classes (including Ambiguous). NEmoSen sentiment Macro-F1 is depressed by near-complete Neutral class failure (135 samples, 12.7 % of corpus).

6 Discussion

6.1 Data Scale and Model Selection (RQ2)

Finding 1: A rank reversal at the data-size boundary. XLM-RoBERTa achieves Micro-F1 = 0.6982 on NepEMO (3,123 training samples) but declines to 0.5391 on NEmoSen (741 training samples)—a drop of 22.8 % relative. By contrast, Linear SVM increases from 0.6166 to 0.6309 (+2.3 % relative), confirming that TF-IDF features generalise robustly to the shorter, more uniform vocabulary of the NEmoSen corpus. This directly answers **RQ2**: for code-mixed Nepali text, TF-IDF + SVM is the robust choice below approximately 750 training samples, whereas transformers require substantially larger corpora to activate their cross-lingual pretraining advantage.

Notably, mBERT (0.5551) outperforms XLM-RoBERTa (0.5391) on NEmoSen despite being architecturally smaller, suggesting that XLM-RoBERTa's optimisation for longer fine-tuning regimes makes it more sensitive to data volume than mBERT.

6.2 Effectiveness of Multi-Head Attention (RQ1)

BiLSTM + MHA consistently outperforms base BiLSTM across both Nepali datasets. On NepEMO the absolute gain is +0.1029 (+21.0 %), reaching 0.5931—within 0.0235 of Linear SVM (0.6166). On NEmoSen the gain is +0.0823 (+19.4 %), reaching 0.5070. This positions attention-enhanced deep learning as a viable computational middle ground between classical ML and full transformer fine-tuning, answering **RQ1** affirmatively.

CNN-LSTM + Attention degrades on NepEMO (−24.0 %). We attribute this to over-parameterisation: the multi-scale convolution front-end already extracts rich n-gram features, and adding a full MHA block introduces redundant capacity that the training set cannot regularise effectively. On NEmoSen this effect is reversed (+3.1 %), consistent with the simpler vocabulary distribution of shorter texts where additional attention capacity provides a modest benefit.

6.3 Challenging Emotion Classes (RQ3)

Depression achieves zero F1 across all models on NEmoSen and near-zero on NepEMO, driven by extreme class imbalance: only approximately 7 Depression test instances exist in the NEmoSen split. The per-class analysis (Table 5) confirms that Joy also underperforms relative to its support, owing to vocabulary overlap with Neutral sentiment posts (expressions of mild positivity are frequently misclassified as Neutral). XLM-RoBERTa achieves the highest per-class F1 on NepEMO across all five emotions (Joy = 0.64, Anger = 0.71, Fear = 0.72, Sadness = 0.75, Depression = 0.62), with Sadness and Fear as the best-predicted categories—consistent with their higher prevalence and more distinctive vocabulary in the training data.

6.4 Sentiment Imbalance

The low Macro-F1 on NEmoSen (best: 0.3874) stems from near-complete Neutral class failure. With only 135 Neutral samples (12.7 %) in a corpus dominated by Negative (56.0 %), all models default to predicting Negative. This warrants SMOTE oversampling or class-weighted loss functions in future work.

6.5 Limitations

All experiments were conducted as single runs with `random_state=42`; no variance estimates are available. The NEmoSen annotation followed a split-partition protocol rather than full dual-annotation, limiting formal inter-annotator agreement reporting. Future work should employ full cross-annotation with reported Cohen's κ to strengthen annotation quality claims.

7 Conclusion

We presented a comprehensive benchmark of ten models for multi-label emotion and sentiment analysis on code-mixed Nepali Reddit text, evaluated across two Nepali corpora (NepEMO, 4,462 samples; NEmoSen [2], 1,059 samples, 2025–2026) and two English benchmarks (MELD, GoEmotions) for cross-lingual generalisation assessment. Key findings are:

1. XLM-RoBERTa achieves state-of-the-art Micro-F1 of 0.6982 on NepEMO, confirming the value of large multilingual pretraining for code-mixed low-resource text at scale.
2. Multi-Head Attention augmentation yields consistent improvements to BiLSTM (+21.0 % on NepEMO; +19.4 % on NEmoSen), positioning it as a computationally efficient alternative to full transformer fine-tuning.
3. For corpora below ≈750 training samples, TF-IDF + Linear SVM is the most reliable choice (Micro-F1 = 0.6309 on NEmoSen), with XLM-RoBERTa showing a rank reversal relative to larger corpora.
4. Depression and Neutral sentiment remain systematically under-predicted due to class imbalance, warranting targeted augmentation strategies.

Future work will explore back-translation data augmentation, combined training across both Nepali datasets, character-level subword features for Devanagari romanisation variability, and full dual-annotation with reported agreement metrics.

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